# MTH166

## Lecture-17

## Method of Undeterminant Coeff.-II

**Problem 4.** Find the general solution of: $y'' + y' - 6y = 39\cos 3x$

**Solution:** The given equation is:

$y'' + y' - 6y = 39\cos 3x$ (1)

S.F. : $(D^2 + D - 6)y = 39\cos 3x$ where $D \equiv \frac{d}{dx}$

$\Rightarrow f(D)y = r(x)$ where $f(D) = (D^2 + D - 6)$ and $r(x) = 39\cos 3x$

To find Complimentary Function (C.F.):

A.E. : $f(D) = 0$ $\Rightarrow (D^2 + D - 6) = 0$ $\Rightarrow (D-2)(D+3) = 0$

$\Rightarrow D = 2, -3$ (real and unequal roots)

$\therefore$ Complimentary function is given by:

$\Rightarrow y_c = c_1 e^{2x} + c_2 e^{-3x}$

To find Particular Integral (P.I.):

Since $r(x) = 39\cos 3x$, So, let trial solution be: $y_p = a\cos 3x + b\sin 3x$

$\Rightarrow y_p{}' = -3a\sin 3x + 3b\cos 3x$ $\Rightarrow y_p{}'' = -9a\cos 3x - 9b\sin 3x$

Since $y_p$ is a solution of equation (1)

So, $y_p{}'' + y_p{}' - 6y_p = 39\cos 3x$

$\Rightarrow (-9a\cos 3x - 9b\sin 3x) + (-3a\sin 3x + 3b\cos 3x) - 6(a\cos 3x + b\sin 3x) = 39\cos 3x$

$\Rightarrow (-15a + 3b)\cos 3x + (-3a - 15b)\sin 3x = 39\cos 3x + 0\sin 3x$

Comparing the like coefficients:

Coeff. of $\cos 3x$: $-15a + 3b = 39$ (2)

Coeff. of $\sin 3x$ : $-3a - 15b = 0$ (3)

Solving equations (2) and (3):

$-15a + 3b = 39$ (2) $\times 3$

$-3a - 15b = 0$ (3) $\times 15$ and subtracting, we get:

$234b = 127 \Rightarrow b = \frac{1}{2}$

Put value of $b$ in equation (3): $3a = -15b = -\frac{15}{2}$ $\Rightarrow a = -\frac{5}{2}$

Putting back values of $a, b$ in $y_p = a\cos 3x + b\sin 3x$

$$y_p = -\frac{5}{2}\cos 3x + \frac{1}{2}\sin 3x$$

General solution is given by: $y =$ C.F. +P.I.

i.e. $y = y_c + y_p$

$$\Rightarrow y = (c_1 e^{2x} + c_2 e^{-3x}) - \frac{5}{2}\cos 3x + \frac{1}{2}\sin 3x$$ **Answer.**

**Problem 5.** Find the general solution of: $y'' + 4y = \cos x$

**Solution:** The given equation is:

$$y'' + 4y = \cos x \qquad (1)$$

S.F. : $(D^2 + 4)y = \cos x$ where $D \equiv \frac{d}{dx}$

$\Rightarrow f(D)y = r(x)$ where $f(D) = (D^2 + 4)$ and $r(x) = \cos x$

To find Complimentary Function (C.F.):

A.E. : $f(D) = 0 \quad \Rightarrow (D^2 + 4) = 0$

$\Rightarrow D = 2i, -2i$ (Complex roots)

$\therefore$ Complimentary function is given by:

$\Rightarrow y_c = c_1 \cos 2x + c_2 \sin 2x$

To find Particular Integral (P.I.):

Since $r(x) = \cos x$, So, let trial solution be: $y_p = a\cos x + b\sin x$

$\Rightarrow y_p{}' = -a\sin x + b\cos x$ $\Rightarrow y_p{}'' = -a\cos x - b\sin x$

Since $y_p$ is a solution of equation (1)

So, $y_p{}'' + 4y_p = \cos x$

$\Rightarrow (-a\cos x - b\sin x) + 4(a\cos x + b\sin x) = \cos x$

$\Rightarrow 3a\cos x + 3b\sin x = \cos x + 0\sin x$

Comparing the like coefficients:

Coeff. of $\cos x$: $3a = 1$ $\Rightarrow a = \frac{1}{3}$

Coeff. of $\sin x$ : $3b = 0$ $\Rightarrow b = 0$

Putting back values of $a, b$ in $y_p = \frac{1}{3}\cos x$

General solution is given by: $y =$ C.F. +P.I.

i.e. $y = y_c + y_p$

$\Rightarrow y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3}\cos x$ **Answer.**

**<u>Problem 6.</u>** Find the general solution of: $y'' + 4y = \cos 2x$

**<u>Solution:</u>** The given equation is:

$$y'' + 4y = \cos x \qquad (1)$$

<u>S.F. :</u> $(D^2 + 4)y = \cos x$ where $D \equiv \frac{d}{dx}$

$\Rightarrow f(D)y = r(x)$ where $f(D) = (D^2 + 4)$ and $r(x) = \cos x$

<u>To find Complimentary Function (C.F.):</u>

<u>A.E. :</u> $f(D) = 0 \quad \Rightarrow (D^2 + 4) = 0$

$\Rightarrow D = 2i, -2i$ (Complex roots)

$\therefore$ Complimentary function is given by:

$\Rightarrow y_c = c_1 \cos 2x + c_2 \sin 2x$

To find Particular Integral (P.I.):

Since $r(x) = \cos 2x$, So, let trial solution be: $y_p = x(a\cos 2x + b\sin 2x)$

$\Rightarrow y_p{}' = x(-2a\sin 2x + 2b\cos 2x) + 1(a\cos 2x + b\sin 2x)$

$\Rightarrow y_p{}' = (a + 2bx)\cos 2x + (b - 2ax)\sin 2x$

$\Rightarrow y_p{}'' = [(a + 2bx)(-2\sin 2x) + \cos 2x\,(2b)] + [(b - 2ax)(2\cos 2x) + \sin 2x(-2a)]$

$\Rightarrow y_p{}'' = (4b - 4ax)\cos 2x - (4a + 4bx)\sin 2x$

Since $y_p$ is a solution of equation (1)

So, $y_p{}'' + 4y_p = \cos 2x$

$\Rightarrow \big((4b - 4ax)\cos 2x - (4a + 4bx)\sin 2x\big) + 4(x(a\cos 2x + b\sin 2x)) = \cos 2x$

$\Rightarrow 4b\cos 2x - 4a\sin 2x = \cos 2x + 0\sin 2x$

Comparing the like coefficients:

Coeff. of $\cos 2x$: $4b = 1$ $\Rightarrow b = \frac{1}{4}$

Coeff. of $\sin 2x$ : $-4a = 0$ $\Rightarrow a = 0$

Putting back values of $a, b$ in $y_p = x(0 \cos 2x + \frac{1}{4}\sin 2x) = \frac{1}{4}x \sin 2x$

General solution is given by: $y =$ C.F. +P.I.

i.e. $y = y_c + y_p$

$\Rightarrow y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4}x \sin 2x$ **Answer.**

## **Polling Quiz**

For $y'' + y = \cos x$, the assumed trial solution will be:

(A) $y_p = x(a\cos x + b\sin x)$

(B) $y_p = (a\cos x + b\sin x)$

(C) $y_p = x(a\cos 2x + b\sin 2x)$

**Problem 7.** Find the general solution of: $y''-4y'+13y=12e^{2x}\sin 3x$

**Solution:** The given equation is:

$$y''-4y'+13y=12e^{2x}\sin 3x \qquad (1)$$

S.F. : $(D^2-4D+13)y=12e^{2x}\sin 3x$ where $D\equiv\frac{d}{dx}$

$\Rightarrow f(D)y=r(x)$ where $f(D)=(D^2-4D+13)$ and $r(x)=12e^{2x}\sin 3x$

To find Complimentary Function (C.F.):

A.E. : $f(D)=0 \quad \Rightarrow (D^2-4D+13)=0$

$\Rightarrow D=2+3i, 2-3i$ (Complex roots)

$\therefore$ Complimentary function is given by:

$\Rightarrow y_c=e^{2x}(c_1\cos 3x+c_2\sin 3x)$

To find Particular Integral (P.I.):

Since $r(x) = 12e^{2x}\sin 3x$, So, let trial solution be: $y_p = xe^{2x}(a\cos 3x + b\sin 3x)$

$\Rightarrow calculate\ y_p{}'\ and y_p{}''$

Since $y_p$ is a solution of equation (1)

So, $y_p{}'' - 4y_p{}' + 13y_p = 12e^{2x}\sin 3x$

Comparing the like coefficients:

$\Rightarrow a = -2$ and $b = 0$

Putting back values of $a, b$ in $y_p = -2xe^{2x}\cos 3x$

General solution is given by: $y =$ C.F. +P.I.

i.e. $y = y_c + y_p$

$\Rightarrow y = e^{2x}(c_1\cos 3x + c_2\sin 3x) - 2xe^{2x}\cos 3x$ **Answer.**